CHAPTER 1: GST (Goods and Services Tax)

Q1. For an intra-state sale, the CGST paid by a dealer to the Central government is ₹120. If the marked price of the article is ₹2000, the rate of GST is: [2024, 1 mark]
(a) 6%
(b) 10%
(c) 12%
(d) 16.67%
Answer: (c) 12%

Q2. The marked price of an article is ₹1375. If the CGST is charged at a rate of 4%, then the price of the article including GST is: [2025 Paper, 1 mark]
(a) ₹55
(b) ₹110
(c) ₹1430
(d) ₹1485
Answer: (d) ₹1485

Q3. The selling price of a shirt excluding GST is ₹800. If the rate of GST is 12%, then the total price of the shirt is: [2022 Sem 1, 1 mark]
(a) ₹704
(b) ₹96
(c) ₹896
(d) ₹848
Answer: (c) ₹896

Q4. The marked price of an article is ₹5000. The shopkeeper gives a discount of 10%. If the rate of GST is 12%, then the amount paid by the customer including GST is: [2022 Sem 1, 2 marks]
(a) ₹5040
(b) ₹6100
(c) ₹6272
(d) ₹6160
Answer: (a) ₹5040

Q5. The printed price of an article is ₹3080. If the rate of GST is 10%, then the GST charged is: [2023, 1 mark]
(a) ₹154

(b) ₹308
(c) ₹30.80
(d) ₹15.40
Answer: (b) ₹308

CHAPTER 2: BANKING (Recurring Deposit Accounts)

Q1. Mr. Anuj deposits ₹500 per month for 18 months in a recurring deposit account at a certain rate. If he earns ₹570 as interest at the time of maturity, then his matured amount is: [2025 Paper, 1 mark]
(a) ₹(500 × 18 + 570)
(b) ₹(500 × 19 + 570)
(c) ₹(500 × 18 × 19 + 570)
(d) ₹(500 × 9 × 19 + 570)
Answer: (a) ₹(500 × 18 + 570)

Q2. Mohit opened a Recurring deposit account in a bank for 2 years. He deposits ₹1000 every month and receives ₹25000 on maturity. The interest he earned in 2 years is: [2022 Sem 1, 1 mark]
(a) ₹13500
(b) ₹3000
(c) ₹24000
(d) ₹1500
Answer: (b) ₹3000

Q3. Naveen deposits ₹800 every month in a recurring deposit account for 6 months. If he receives ₹4884 at the time of maturity, then the interest he earns is: [2023, 1 mark]
(a) ₹84
(b) ₹42
(c) ₹24
(d) ₹284
Answer: (a) ₹84

Q4. Joseph has a recurring deposit account in a bank for two years at the rate of 8% per annum simple interest. If at the time of maturity Joseph receives ₹2000 as interest, then the monthly installment is: [2022 Sem 1, 1 mark]
(a) ₹1200

(b) ₹600
(c) ₹1000
(d) ₹1600
Answer: (c) ₹1000

Q5. For the above question, the total amount deposited in the bank is: [2022 Sem 1, 1 mark]
(a) ₹25000
(b) ₹24000
(c) ₹26000
(d) ₹23000
Answer: (b) ₹24000

Q6. Mr. Gupta opened a recurring deposit account in a bank. He deposited Rs. 2500 per month for two years. At the time of maturity he got Rs. 67,500. The total interest earned by Mr. Gupta is: [2010, 4 marks]
(a) ₹60,000
(b) ₹7,500
(c) ₹67,500
(d) ₹5,000
Answer: (b) ₹7,500

Q7. Sonia had a recurring deposit account in a bank and deposited ₹600 per month for 2½ years. If the rate of interest was 10% p.a., the maturity value of this account is: [2018, 3 marks]
(a) ₹18,000
(b) ₹20,325
(c) ₹21,000
(d) ₹22,500
Answer: (b) ₹20,325

Q8. Ahmed has a recurring deposit account in a bank. He deposits Rs. 2,500 per month for 2 years. If he gets Rs. 66,250 at the time of maturity, the interest paid by the bank is: [2011, 3 marks]
(a) ₹60,000
(b) ₹6,250
(c) ₹66,250
(d) ₹10,000
Answer: (b) ₹6,250

Q9. Kiran deposited Rs. 200 per month for 36 months in a bank's recurring deposit account. If the bank pays interest at the rate of 11% per annum, the amount she gets on maturity is: [2012, 3 marks]
(a) ₹7,200
(b) ₹8,400
(c) ₹9,000
(d) ₹9,621
Answer: (d) ₹9,621

Q10. Shahrukh opened a 'Recurring Deposit' account in a bank and deposited Rs. 800 per month for 1½ years. If he received Rs. 15,084 at the time of maturity, the rate of interest per annum is: [2014, 3 marks]
(a) 6%
(b) 8%
(c) 10%
(d) 12%
Answer: (b) 8%

Q11. Katrina opened a recurring deposit account with a Nationalised Bank for a period of 2 years. If the bank pays interest at the rate 6% per annum and the monthly instalment is Rs. 1,000, the interest earned in 2 years is: [2015, 3 marks]
(a) ₹1,200
(b) ₹1,500
(c) ₹1,800
(d) ₹2,000
Answer: (b) ₹1,500

Q12. Mohan has a recurring deposit account in a bank for 2 years at 6% p.a. simple interest. If he gets Rs. 1200 as interest at the time of maturity, the monthly installment is: [2016, 3 marks]
(a) ₹600
(b) ₹700
(c) ₹800
(d) ₹900
Answer: (c) ₹800

Q13. Mrs. Goswami deposits 1000 every month in a recurring deposit account for 3 years at 8% interest per annum. The matured value is: [2009, 3 marks]
(a) ₹36,000
(b) ₹40,440

(c) ₹44,000
(d) ₹50,000
Answer: (b) ₹40,440

Q14. Mr. Richard has a recurring deposit account in a bank for 3 years at 7.5% p. a. simple interest. If he gets Rs. 8325 as interest at the time of maturity, the monthly deposit is: [2017, 4 marks]
(a) ₹1,500
(b) ₹1,800
(c) ₹2,000
(d) ₹2,500
Answer: (c) ₹2,000

CHAPTER 3: SHARES AND DIVIDENDS

Q1. The sum invested to purchase 15 shares of a company of nominal value ₹75 available at a discount of 20% is: [2024, 1 mark]
(a) ₹60
(b) ₹90
(c) ₹1350
(d) ₹900
Answer: (d) ₹900

Q2. A man invested in a company paying 12% dividend on its share. If the percentage return on his investment is 10%, then the shares are: [2025 Paper, 1 mark]
(a) at par
(b) below par
(c) above par
(d) cannot be determined
Answer: (c) above par

Q3. A company with 500 shares of nominal value ₹120 declares an annual dividend of 15%. The total amount of dividend paid by the company is: [2020, 3 marks]
(a) ₹7,200
(b) ₹9,000
(c) ₹10,800

(d) ₹12,000
Answer: (b) ₹9,000

Q4. A man invests Rs. 9,600 on Rs. 100 shares at Rs. 20 premium paying 8% dividend. The original number of shares is: [2015, 4 marks]
(a) 80
(b) 96
(c) 100
(d) 120
Answer: (a) 80

Q5. A man invests Rs. 22,500 in Rs. 50 shares available at 10% discount. If the dividend paid by the company is 12%, the number of shares purchased is: [2018, 3 marks]
(a) 400
(b) 450
(c) 500
(d) 550
Answer: (c) 500

Q6. Salman buys 50 shares of face value Rs. 100 available at Rs. 132. His investment is: [2013, 4 marks]
(a) ₹5,000
(b) ₹6,600
(c) ₹13,200
(d) ₹50,000
Answer: (b) ₹6,600

Q7. A man bought 200 shares of a company at 25% premium. If he received a return of 5% on his investment, the market value is: [2025 Paper, 3 marks]
(a) ₹200
(b) ₹225
(c) ₹250
(d) ₹275
Answer: (c) ₹250

Q8. Amit Kumar invests 36,000 in buying 100 shares at 20 premium. The number of shares he buys is: [2009, 3 marks]
(a) 200
(b) 250
(c) 300

(d) 360
Answer: (c) 300

Q9. Mr. Parekh invested Rs. 52,000 on Rs. 100 shares at a discount of Rs. 20 paying 8% dividend. The annual dividend is: [2011, 3 marks]
(a) ₹4,000
(b) ₹4,160
(c) ₹5,200
(d) ₹6,000
Answer: (c) ₹5,200

Q10. A company with 4000 shares of nominal value of 110 each declares an annual dividend of 15%. The total amount of dividend paid by the company is: [2008, 4 marks]
(a) ₹60,000
(b) ₹66,000
(c) ₹70,000
(d) ₹75,000
Answer: (b) ₹66,000

CHAPTER 4: LINEAR INEQUATIONS

Q1. The solution set of the inequation $x-3 \geq -5$, $x \in R$ is: [2022 Sem 1, 1 mark]
(a) {x : x > -2, x ∈ R}
(b) {x : x ≤ -2, x ∈ R}
(c) {x : x ≥ -2, x ∈ R}
(d) {-2, -1, 0, 1, 2}
Answer: (c) {x : x ≥ -2, x ∈ R}

Q2. The solution set for $0 < -x/3 < 2$, $x \in Z$ is: [2025 Paper, 1 mark]
(a) {-5, -4, -3, -2, -1}
(b) {-6, -5, -4, -3, -2, -1}
(c) {-5, -4, -3, -2, -1, 0}
(d) {-6, -5, -4, -3, -2, -1, 0}
Answer: (a) {-5, -4, -3, -2, -1}

Q3. The solution set for the inequation $2x+4 \leq 14$, $x \in W$ is: [2023, 1 mark]
(a) {1, 2, 3, 4, 5}
(b) {0, 1, 2, 3, 4, 5}

(c) {1, 2, 3, 4}
(d) {0, 1, 2, 3, 4}
Answer: (b) {0, 1, 2, 3, 4, 5}

Q4. Solve the inequation and represent the solution set on the number line: $2x - 5/3 < (3x)/5 + 10 \leq (4x)/5 + 11$, $x \in R$. The solution set is: [2025 Paper, 3 marks]
(a) $-5 \leq x < 8\frac{1}{3}$
(b) $-5 < x \leq 8\frac{1}{3}$
(c) $-5 \leq x \leq 8\frac{1}{3}$
(d) $-5 < x < 8\frac{1}{3}$
Answer: (a) $-5 \leq x < 8\frac{1}{3}$

Q5. Solve the following inequation: $-2 + 10x \leq 13x + 10 < 24 + 10x$, $x \in Z$. The solution set is: [2018, 3 marks]
(a) {-4, -3, -2, -1, 0, 1, 2, 3, 4}
(b) {-4, -3, -2, -1, 0, 1, 2, 3}
(c) {-4, -3, -2, -1, 0, 1, 2}
(d) {-4, -3, -2, -1, 0, 1}
Answer: (a) {-4, -3, -2, -1, 0, 1, 2, 3, 4}

Q6. Solve the inequation: $4x - 19 < (3x)/5 - 2 \leq -2/5 + x$, $x \in R$. The solution set is: [2012, 3 marks]
(a) $x < 5$
(b) $x \leq 5$
(c) $x > -5$
(d) $-5 < x < 5$
Answer: (d) $-5 < x < 5$

Q7. Solve the inequation: $-8\frac{1}{2} < -\frac{1}{2} - 4x \leq 7\frac{1}{2}$, $x \in I$. The solution set is: [2017, 3 marks]
(a) {-2, -1, 0, 1}
(b) {-2, -1, 0, 1, 2}
(c) {-1, 0, 1}
(d) {-2, -1, 0}
Answer: (a) {-2, -1, 0, 1}

CHAPTER 5: QUADRATIC EQUATIONS

Q1. The roots of the quadratic equation $px^2 - qx + r = 0$ are real and equal if: [2024, 1 mark]
(a) $p^2 = 4qr$
(b) $q^2 = 4pr$
(c) $-q^2 = 4pr$
(d) $p^2 > 4qr$
Answer: (b) $q^2 = 4pr$

Q2. The given quadratic equation $3x^2 + \sqrt{7}x + 2 = 0$ has: [2025 Paper, 1 mark]
(a) two equal real roots
(b) two distinct real roots
(c) more than two real roots
(d) no real roots
Answer: (d) no real roots

Q3. Which of the following quadratic equations has 2 and 3 as its roots? [2022 Sem 1, 1 mark]
(a) $x^2 - 5x + 6 = 0$
(b) $x^2 + 5x + 6 = 0$
(c) $x^2 - 5x - 6 = 0$
(d) $x^2 + 5x - 6 = 0$
Answer: (a) $x^2 - 5x + 6 = 0$

Q4. If 3 is a root of the quadratic equation $x^2 - px + 3 = 0$, then p is equal to: [2023, 1 mark]
(a) 4
(b) 5
(c) 6
(d) 7
Answer: (a) 4

Q5. One of the roots of the quadratic equation $x^2 - 8x + 5 = 0$ is 7.3166. The root of the equation correct to 4 significant figures is: [2022 Sem 1, 1 mark]
(a) 7.3166
(b) 7.317
(c) 7.316
(d) 7.32
Answer: (b) 7.317

Q6. Solve the quadratic equation $2x^2 - 10x + 5 = 0$ correct up to three significant figures. The roots are: [2024, 3 marks]
(a) 4.43, 0.563
(b) 4.5, 0.56
(c) 4.4, 0.56
(d) 4.43, 0.56
Answer: (a) 4.43, 0.563

Q7. Solve the quadratic equation $x^2 - 7x + 3 = 0$ correct to two decimal places. The roots are: [2020, 3 marks]
(a) 6.54, 0.46
(b) 6.5, 0.5
(c) 6.54, 0.5
(d) 6.5, 0.46
Answer: (a) 6.54, 0.46

Q8. Solve the quadratic equation $2x^2 - 5x - 4 = 0$ correct to three significant figures. The roots are: [2025 Paper, 4 marks]
(a) 3.14, -0.637
(b) 3.14, -0.64
(c) 3.1, -0.64
(d) 3.14, -0.63
Answer: (a) 3.14, -0.637

Q9. Solve the quadratic equation $x^2 + 4x - 8 = 0$ correct to one decimal place. The roots are: [2023, 4 marks]
(a) 1.5, -5.5
(b) 1.4, -5.4
(c) 1.5, -5.4
(d) 1.4, -5.5
Answer: (a) 1.5, -5.5

Q10. Without solving the following quadratic equation, find the value of 'p' for which the roots are equal: $px^2 - 4x + 3 = 0$. The value of p is: [2010, 3 marks]
(a) 3/4
(b) 4/3
(c) 1
(d) 2
Answer: (b) 4/3

Q11. Without solving the following quadratic equation, find the value of 'p' for which the given equation has real and equal roots: $x^2 + (p-3)x + p = 0$. The value of p is: [2013, 4 marks]
(a) 1, 9
(b) 1, 8
(c) 1, 7
(d) 1, 6
Answer: (a) 1, 9

Q12. Without solving the following quadratic equation, find the value of 'm' for which the given equation has real and equal roots: $x^2 + 2(m-1)x + (m+5) = 0$. The value of m is: [2012, 4 marks]
(a) -1, 5
(b) 1, -5
(c) -1, -5
(d) 1, 5
Answer: (a) -1, 5

Q13. Solve the following quadratic equation and give your answer correct to two significant figures: $4x^2 - 7x + 2 = 0$. The roots are: [2009, 3 marks]
(a) 1.4, 0.36
(b) 1.4, 0.4
(c) 1.5, 0.36
(d) 1.4, 0.35
Answer: (a) 1.4, 0.36

CHAPTER 6: FACTORISATION (REMAINDER AND FACTOR THEOREM)

Q1. What must be subtracted from the polynomial $x^3 + x^2 - 2x + 1$, so that the result is exactly divisible by $(x-3)$? [2024, 1 mark]
(a) -31
(b) -30
(c) 30
(d) 31
Answer: (d) 31

Q2. The factor common to the two polynomials $x^2 - 4$ and $x^3 - x^2 - 4x + 4$ is: [2025 Paper, 1 mark]
(a) (x + 1)
(b) (x - 1)
(c) (x + 2)
(d) (x - 2)
Answer: (c) (x + 2) and (d) (x - 2) [Note: Both are correct]

Q3. If (x + 2) is a factor of the polynomial $x^3 - kx^2 - 5x + 6$, then the value of k is: [2022 Sem 1, 1 mark]
(a) 1
(b) 2
(c) 3
(d) -2
Answer: (b) 2

Q4. The polynomial $x^3 - 2x^2 + ax + 12$ when divided by (x + 1) leaves remainder 20, then 'a' is equal to: [2022 Sem 1, 1 mark]
(a) -31
(b) 9
(c) 11
(d) -11
Answer: (d) -11

Q5. (x + 2) and (x + 3) are two factors of the polynomial $x^3 + 6x^2 + 11x + 6$. If this polynomial is completely factorised the result is: [2022 Sem 1, 2 marks]
(a) (x - 2)(x + 3)(x + 1)
(b) (x + 2)(x - 3)(x - 1)
(c) (x + 2)(x + 3)(x - 1)
(d) (x + 2)(x + 3)(x + 1)
Answer: (d) (x + 2)(x + 3)(x + 1)

Q6. If $x-2$ is a factor of $x^3 - kx - 12$, then the value of k is: [2023, 1 mark]
(a) 3
(b) 2
(c) -2
(d) -3
Answer: (c) -2

Q7. When divided by $x-3$ the polynomials $x^3 - px^2 + x + 6$ and $2x^3 - x^2 - (p+3)x - 6$ leave the same remainder. The value of 'p' is: [2010, 3 marks]
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (a) 1

Q8. Using the Remainder Theorem factorise completely the following polynomial: $2x^3 + x^2 - 13x + 6$. The factors are: [2010, 4 marks]
(a) (x - 2)(x + 3)(2x - 1)
(b) (x - 2)(x - 3)(2x + 1)
(c) (x + 2)(x - 3)(2x + 1)
(d) (x + 2)(x + 3)(2x - 1)
Answer: (b) (x - 2)(x - 3)(2x + 1)

Q9. Find the value of 'k' if (x - 2) is a factor of $x^3 + 2x^2 - kx + 10$. The value of k is: [2011, 3 marks]
(a) 13
(b) -13
(c) 10
(d) -10
Answer: (a) 13

Q10. Using the Remainder Theorem factorise completely the following polynomial: $3x^3 + 2x^2 - 19x + 6$. The factors are: [2012, 3 marks]
(a) (x - 2)(3x - 1)(x + 3)
(b) (x - 2)(3x + 1)(x - 3)
(c) (x + 2)(3x - 1)(x - 3)
(d) (x + 2)(3x + 1)(x + 3)
Answer: (a) (x - 2)(3x - 1)(x + 3)

Q11. If (x - 2) is a factor of the expression $2x^3 + ax^2 + bx - 14$ and when the expression is divided by (x - 3), it leaves a remainder 52, the values of a and b are: [2013, 3 marks]
(a) a = 5, b = 2
(b) a = 5, b = -2
(c) a = -5, b = 2
(d) a = -5, b = -2
Answer: (b) a = 5, b = -2

Q12. Using the Remainder and Factor Theorem, factorise the following polynomial: $x^3 + 10x^2 - 37x + 26$. The factors are: [2014, 3 marks]
(a) (x - 1)(x + 2)(x + 13)
(b) (x - 1)(x - 2)(x + 13)
(c) (x + 1)(x + 2)(x - 13)
(d) (x + 1)(x - 2)(x - 13)
Answer: (b) (x - 1)(x - 2)(x + 13)

Q13. Find 'a' of the two polynomials $ax^3 + 3x^2 - 9$ and $2x^3 + 4x + a$, leaves the same remainder when divided by $x+3$. The value of a is: [2015, 3 marks]
(a) -24
(b) 24
(c) -12
(d) 12
Answer: (a) -24

Q14. Using remainder theorem, find the value of k if on dividing $2x^3 + 3x^2 - kx + 5$ by $x-2$, leaves a remainder 7. The value of k is: [2016, 3 marks]
(a) 8
(b) 9
(c) 10
(d) 11
Answer: (c) 10

Q15. If $(x-2)$ is a factor of $2x^3 - x^2 - px - 2$, the value of p is: [2008, 4 marks]
(a) 4
(b) 5
(c) 6
(d) 7
Answer: (b) 5

Q16. Given that $x+2$ and $x+3$ are factors of $2x^3 + ax^2 + 7x - b$. The values of a and b are: [2009, 4 marks]
(a) a = 9, b = 6
(b) a = 9, b = -6
(c) a = -9, b = 6
(d) a = -9, b = -6
Answer: (a) a = 9, b = 6

Q17. What must be subtracted from $16x^3 - 8x^2 + 4x + 7$ so that the resulting expression has $2x+1$ as a factor? [2017, 3 marks]
(a) 0
(b) 1
(c) 2
(d) 3
Answer: (b) 1

CHAPTER 7: MATRICES

Q1. If matrix $A = [[2,2],[0,2]]$ and $A^2 = [[4,x],[0,4]]$, then the value of x is: [2024, 1 mark]
(a) 2
(b) 4
(c) 8
(d) 10
Answer: (c) 8

Q2. If $A = [[0,1],[1,0]]$, then A^2 is equal to: [2025 Paper, 1 mark]
(a) $[[1,0],[1,0]]$
(b) $[[0,1],[0,1]]$
(c) $[[1,0],[0,1]]$
(d) $[[0,1],[1,0]]$
Answer: (c) $[[1,0],[0,1]]$

Q3. The product AB of two matrices A and B is possible if: [2022 Sem 1, 1 mark]
(a) A and B have the same number of rows
(b) the number of columns of A is equal to the number of rows of B
(c) the number of rows of A is equal to the number of columns of B
(d) A and B have the same number of columns
Answer: (b) the number of columns of A is equal to the number of rows of B

Q4. If $A = [[5,10],[3,-4]]$ and $I = [[1,0],[0,1]]$, then AI is equal to: [2022 Sem 1, 1 mark]
(a) $[[1,0],[0,1]]$
(b) $[[5,10],[-3,4]]$
(c) $[[5,10],[3,-4]]$

(d) [[15,15],[-1,-1]]
Answer: (c) [[5,10],[3,-4]]

Q5. Given A = [[1,3],[2,4]], B = [[1,2],[2,4]], C = [[4,1],[1,5]] and I = [[1,0],[0,1]]. A(B+C)−14I is equal to: [2023, 4 marks]
(a) [[0,30],[22,28]]
Answer: (a) [[0,30],[22,28]]

Q6. Given [[2,1],[-3,4]] X = [[7],[6]]. The matrix X is: [2012, 3 marks]
(a) [[2],[3]]
(b) [[3],[2]]
(c) [[4],[1]]
(d) [[1],[4]]
Answer: (a) [[2],[3]]

Q7. If A = [[3,1],[-1,2]] and I = [[1,0],[0,1]], then A^2 − 5A + 7I is equal to: [2012, 3 marks]
(a) [[0,0],[0,0]]
(b) [[1,0],[0,1]]
(c) [[0,1],[1,0]]
(d) [[1,1],[1,1]]
Answer: (a) [[0,0],[0,0]]

Q8. Given A = [[2,-6],[2,0]], B = [[-3,2],[4,0]], C = [[4,0],[0,2]]. The matrix X such that A + 2X = 2B + C is: [2013, 3 marks]
(a) [[-2,2],[1,1]]
(b) [[-2,2],[1,-1]]
(c) [[-2,2],[-1,1]]
(d) [[2,2],[1,1]]
Answer: (c) [[-2,2],[-1,1]]

Q9. Let A = [[2,1],[0,-2]], B = [[4,1],[-3,-2]] and C = [[-3,2],[-1,4]]. A^2 + AC − 5B is equal to: [2014, 3 marks]
(a) [[-3,0],[18,0]]
Answer: (a) [[-3,0],[18,0]]

Q10. Given matrix B = [[1,1],[8,3]], find the matrix X if, X = B^2 − 4B. X is equal to: [2017, 3 marks]
(a) [[-2,0],[16,-2]]
(b) [[2,0],[-16,2]]

(c) `[[-2,0],[16,2]]`
(d) `[[2,0],[16,2]]`
Answer: (a) `[[-2,0],[16,-2]]`

Q11. Given `A = [[3,5],[4,-2]]` and `B = [[2],[4]]`. The product AB is: [2011, 3 marks]
(a) `[[26],[0]]`
Answer: (a) `[[26],[0]]`

CHAPTER 8: ARITHMETIC PROGRESSION

Q1. The 7th term of the Arithmetic Progression `1/a, (1/a + 1), (1/a + 2),...` is: [2024, 1 mark]
(a) `1/a + 6`
(b) `1/a + 7`
(c) `1/a + 8`
(d) `1/a + 9`
Answer: (a) `1/a + 6`

Q2. If 70, 75, 80, 85 are the first four terms of an Arithmetic Progression, then the 10th term is: [2022 Sem 1, 1 mark]
(a) 35
(b) 25
(c) 115
(d) 105
Answer: (c) 115

Q3. In an Arithmetic Progression if first term is 5, common difference is -3 and the nth term is -7, then n is equal to: [2022 Sem 1, 1 mark]
(a) 5
(b) 17
(c) -13
(d) 7
Answer: (a) 5

Q4. The first four terms of an Arithmetic Progression whose first term is 4 and common difference is -6 are: [2022 Sem 1, 1 mark]
(a) 4, -10, -16, -22

(b) 4, 10, 16, 22
(c) 4, -2, -8, -14
(d) 4, 2, 8, 14
Answer: (c) 4, -2, -8, -14

Q5. The sum of the first 20 terms of the Arithmetic Progression 2, 4, 6, 8, ... is: [2022 Sem 1, 2 marks]
(a) 400
(b) 840
(c) 420
(d) 800
Answer: (c) 420

Q6. The nth term of an Arithmetic Progression is $2n + 5$. The 10th term is: [2023, 1 mark]
(a) 7
(b) 15
(c) 25
(d) 45
Answer: (c) 25

Q7. The nth term of an Arithmetic Progression (A.P.) is given by the relation $T = 6(7-n)$. The first term and common difference are: [2024, 3 marks]
(a) 36, -6
(b) 36, 6
(c) 42, -6
(d) 42, 6
Answer: (a) 36, -6

Q8. If $(k-3)$, $(2k+1)$ and $(4k+3)$ are three consecutive terms of an A.P., the value of k is: [2018, 3 marks]
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (b) 2

Q9. The first term of an Arithmetic Progression (A.P.) is 5, the last term is 50 and their sum is 440. The number of terms is: [2025 Paper, 3 marks]
(a) 14

(b) 15
(c) 16
(d) 17
Answer: (c) 16

Q10. In an Arithmetic Progression (A.P.) the fourth and sixth terms are 8 and 14 respectively. The first term and common difference are: [2019, 4 marks]
(a) a = 2, d = 3
(b) a = 2, d = 2
(c) a = 2, d = 4
(d) a = 3, d = 2
Answer: (b) a = 2, d = 2

Q11. The sum of the first three terms of an Arithmetic Progression (A.P.) is 42 and the product of the first and third term is 52. The first term and common difference are: [2019, 3 marks]
(a) 14, ±12
(b) 14, ±10
(c) 12, ±14
(d) 12, ±12
Answer: (a) 14, ±12

Q12. If the 6th term of an A.P. is equal to four times its first term and the sum of first six terms is 75, the first term and common difference are: [2020, 3 marks]
(a) a = 5, d = 3
(b) a = 5, d = 4
(c) a = 3, d = 5
(d) a = 4, d = 5
Answer: (a) a = 5, d = 3

Q13. Which term of the Arithmetic Progression (A.P.) 15, 30, 45, 60 ... is 300? [2019, 3 marks]
(a) 18th term
(b) 19th term
(c) 20th term
(d) 21st term
Answer: (c) 20th term

Q14. The 4th term of an A.P. is 22 and 15th term is 66. The sum of the series to 8 terms is: [2018, 4 marks]

(a) 190
(b) 192
(c) 194
(d) 196
Answer: (b) 192

CHAPTER 9: GEOMETRIC PROGRESSION

Q1. 15, 30, 60, 120... are in G.P. The nth term of this G.P. in terms of n is: [2024, 2 marks]
(a) $15 \times 2^{n-1}$
(b) 15×2^{n}
(c) $15 \times 2^{n+1}$
(d) $15 \times 2^{n-2}$
Answer: (a) $15 \times 2^{n-1}$

Q2. The 4th term of a G.P. is 16 and the 7th term is 128. The first term and common ratio are: [2018, 3 marks]
(a) a = 2, r = 2
(b) a = 2, r = 4
(c) a = 4, r = 2
(d) a = 4, r = 4
Answer: (a) a = 2, r = 2

Q3. The 4th, 6th and the last term of a geometric progression are 10, 40 and 640 respectively. If the common ratio is positive, the first term and common ratio are: [2020, 4 marks]
(a) a = 5/4, r = 2
(b) a = 5/4, r = 4
(c) a = 5/2, r = 2
(d) a = 5/2, r = 4
Answer: (a) a = 5/4, r = 2

Q4. The first and last term of a Geometrical Progression (G.P.) are 3 and 96 respectively. If the common ratio is 2, the number of terms is: [2019, 3 marks]
(a) 5
(b) 6
(c) 7

(d) 8
Answer: (b) 6

Q5. If 1701 is the nth term of the Geometric Progression (G.P.) 7, 21, 63 .... the value of n is: [2025 Paper, 3 marks]
(a) 5
(b) 6
(c) 7
(d) 8
Answer: (b) 6

CHAPTER 10: RATIO AND PROPORTION

Q1. If `x, 5.4, 5, 9` are in proportion, then x is: [2022 Sem 1, 1 mark]
(a) 3
(b) 9.72
(c) 25
(d) 25/3
Answer: (a) 3

Q2. If a, b, c, and d are proportional, then `(a+b)/(a-b)` is equal to: [2022 Sem 1, 1 mark]
(a) `c/d`
(b) `(c-d)/(c+d)`
(c) `d/c`
(d) `(c+d)/(c-d)`
Answer: (d) `(c+d)/(c-d)`

Q3. If x, y, z are in continued proportion, then $(y^2+z^2):(x^2+y^2)$ is equal to: [2022 Sem 1, 2 marks]
(a) z : x
(b) x : z
(c) zx
(d) (y + z) : (x + y)
Answer: (a) z : x

Q4. The mean proportional between 4 and 9 is: [2023, 1 mark]
(a) 4
(b) 6

(c) 9
(d) 36
Answer: (b) 6

Q5. If $(a+b)^3/(a-b)^3 = 64/27$, then $a:b$ is: [2024, 4 marks]
(a) 4 : 3
(b) 3 : 4
(c) 7 : 1
(d) 1 : 7
Answer: (c) 7 : 1

Q6. What least number must be added to each of the numbers 5, 11, 19 and 37 so that they are in proportion? [2009, 3 marks]
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (b) 2

Q7. If $(x^2+y^2)/(x^2-y^2) = 17/8$, then the value of x : y is: [2014, 3 marks]
(a) 5 : 3
(b) 3 : 5
(c) 5 : 4
(d) 4 : 5
Answer: (a) 5 : 3

Q8. If $(x-9):(3x+6)$ is the duplicate ratio of $4:9$, the value of x is: [2014, 3 marks]
(a) 5
(b) 6
(c) 7
(d) 8
Answer: (a) 5

Q9. If $8a-5b/8c-5d = 8a+5b/8c+5d$, then $a/b = c/d$. This is proved by which property? [2008, 3 marks]
(a) Componendo
(b) Dividendo
(c) Alternendo
(d) Componendo and Dividendo
Answer: (c) Alternendo

Q10. If $(x^4+1)/(2x^2) = 17/8$, the value of x is: [2013, 3 marks]
(a) 2, 1/2
(b) 2, -1/2
(c) -2, 1/2
(d) -2, -1/2
Answer: (a) 2, 1/2

Q11. 6 is the mean proportion between two numbers x and y and 48 is the third proportional of x and y. The numbers are: [2011, 3 marks]
(a) 3, 12
(b) 4, 9
(c) 6, 6
(d) 2, 18
Answer: (a) 3, 12

CHAPTER 11: COORDINATE GEOMETRY

Q1. Points A(x, y), B(3, -2) and C(4, -5) are collinear. The value of y in terms of x is: [2024, 1 mark]
(a) 3x - 11
(b) 11 - 3x
(c) 3x - 7
(d) 7 - 3x
Answer: (d) 7 - 3x

Q2. The coordinates of the point P(-3, 5) on reflecting on the X axis are: [2022 Sem 2, 1 mark]
(a) (3, 5)
(b) (-3, -5)
(c) (3, -5)
(d) (-3, 5)
Answer: (b) (-3, -5)

Q3. A(1, 4), B(4, 1) and C(x, 4) are the vertices of ΔABC. If the centroid of the triangle is G(4, 3), then x is equal to: [2022 Sem 2, 1 mark]
(a) 2
(b) 1

(c) 7
(d) 4
Answer: (c) 7

Q4. The coordinates of the vertices of ΔABC are respectively (-4, -2), (6, 2) and (4, 6). The centroid G of ΔABC is: [2023, 1 mark]
(a) (2, 2)
(b) (2, 3)
(c) (3, 3)
(d) (0, -1)
Answer: (a) (2, 2)

Q5. P(5, 3) and Q(-5, 3) are divided by the y-axis in the ratio: [2020, 4 marks]
(a) 1 : 2
(b) 2 : 1
(c) 1 : 1
(d) 3 : 2
Answer: (c) 1 : 1

Q6. The line through A(-2, 3) and B(4, b) is perpendicular to the line $2x-4y=5$. The value of b is: [2012, 3 marks]
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (d) 4

Q7. A and B are two points on the x-axis and y-axis respectively. P(2, -3) is the mid-point of AB. The coordinates of A and B are: [2010, 4 marks]
(a) A(4, 0), B(0, -6)
(b) A(4, 0), B(0, 6)
(c) A(-4, 0), B(0, -6)
(d) A(-4, 0), B(0, 6)
Answer: (a) A(4, 0), B(0, -6)

Q8. P(1, -2) is a point on the line segment A(3, -6) and B(x, y) such that AP : PB = 2 : 3. The coordinates of B are: [2017, 3 marks]
(a) (-2, 4)
(b) (2, -4)
(c) (-2, -4)

(d) (2, 4)
Answer: (a) (-2, 4)

Q9. AB is a diameter of a circle with centre C = (-2, 5). If A = (3, -7), the coordinates of B are: [2013, 4 marks]
(a) (-7, 17)
(b) (7, -17)
(c) (-7, -17)
(d) (7, 17)
Answer: (a) (-7, 17)

Q10. Three vertices of a parallelogram ABCD taken in order are A(3, 6), B(5, 10) and C(3, 2). The coordinate of the fourth vertex D is: [2015, 3 marks]
(a) (1, -2)
(b) (-1, 2)
(c) (1, 2)
(d) (-1, -2)
Answer: (a) (1, -2)

CHAPTER 12: COORDINATE GEOMETRY (SLOPE & EQUATION OF LINE)

Q1. The equation of the line passing through origin and parallel to the line $3x+4y+7=0$ is: [2025 Paper, 1 mark]
(a) $3x+4y+5=0$
(b) $4x-3y-5=0$
(c) $4x-3y=0$
(d) $3x+4y=0$
Answer: (d) $3x+4y=0$

Q2. If two lines are perpendicular to one another, then the relation between their slopes m_1 and m_2 is: [2022 Sem 2, 1 mark]
(a) $m_1 = m_2$
(b) $m_1 = 1/m_2$
(c) $m_1 = -m_2$
(d) $m_1 \times m_2 = -1$
Answer: (d) $m_1 \times m_2 = -1$

Q3. The slope of a line joining P(6, k) and Q(1 - 3k, 3) is $1/2$. The value of k is: [2016, 3 marks]
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (d) 4

Q4. Find the value of p for which the lines $2x+3y-7=0$ and $4y-px-12=0$ are perpendicular to each other. The value of p is: [2009, 3 marks]
(a) 4
(b) 5
(c) 6
(d) 7
Answer: (c) 6

Q5. Find the equation of a line with x-intercept = 5 and passing through the point (4, -7). The equation is: [2009, 3 marks]
(a) 7x - y - 35 = 0
(b) 7x + y - 35 = 0
(c) 7x - y + 35 = 0
(d) 7x + y + 35 = 0
Answer: (a) 7x - y - 35 = 0

Q6. Points A and B have coordinates (7, -3) and (1, 9) respectively. The slope of AB is: [2008, 3 marks]
(a) -2
(b) 2
(c) -1
(d) 1
Answer: (a) -2

Q7. For the above question, the equation of the perpendicular bisector of the line segment AB is: [2008, 3 marks]
(a) x - 2y + 2 = 0
(b) x + 2y + 2 = 0
(c) x - 2y - 2 = 0
(d) x + 2y - 2 = 0
Answer: (a) x - 2y + 2 = 0

CHAPTER 13: SIMILARITY

Q1. In the given figure, AB = 24 cm, AC = 18 cm, DE = 12 cm, DF = 9 cm and ∠BAC = ∠EDF. Then ΔABC ~ ΔDEF by the condition: [2022 Sem 1, 1 mark]
(a) SSS
(b) SAS
(c) AAA
(d) RHS
Answer: (b) SAS

Q2. In the given figure, PQ is parallel to TR, then by using condition of similarity: [2022 Sem 1, 1 mark]
(a) `PQ/RT = OP/OT = OQ/OR`
(b) `PQ/RT = OP/OR = OQ/OT`
(c) `PQ/RT = OR/OP = OQ/OT`
(d) `PQ/RT = OP/OR = OT/OQ`
Answer: (b) `PQ/RT = OP/OR = OQ/OT`

Q3. In the given figure ABCD is a trapezium in which DC is parallel to AB. AB = 16 cm and DC = 8 cm. The pair of similar triangles is: [2022 Sem 1, 1 mark]
(a) ΔOAB, ΔOBC
(b) ΔCOD, ΔAOB
(c) ΔADB, ΔACB
(d) ΔCOD, ΔCOB
Answer: (b) ΔCOD, ΔAOB

Q4. For the above question, the ratio of the sides of the pair of similar triangles is: [2022 Sem 1, 1 mark]
(a) 1 : 3
(b) 1 : 2
(c) 2 : 3
(d) 3 : 1
Answer: (b) 1 : 2

Q5. In the given figure, ∠BAP = ∠DCP = 70°, PC = 6 cm and CA = 4 cm, then PD : DB is: [2023, 1 mark]
(a) 5 : 3

(b) 3 : 5
(c) 3 : 2
(d) 2 : 3
Answer: (c) 3 : 2

Q6. In ΔABC, ∠ABC = 90°, AB = 20 cm, AC = 25 cm, DE ⊥ AC and DE = 3 cm. ΔABC ~ ΔAED. The length of BC is: [2025 Paper, 4 marks]
(a) 10 cm
(b) 12 cm
(c) 15 cm
(d) 18 cm
Answer: (c) 15 cm

Q7. In the given diagram ΔADB and ΔACB are two right angled triangles with ∠ADB = ∠BCA = 90°. If AB = 10 cm, AD = 6 cm, BC = 2.4 cm and DP = 4.5 cm, the length of PB is: [2024, 4 marks]
(a) 2 cm
(b) 3 cm
(c) 4 cm
(d) 5 cm
Answer: (b) 3 cm

Q8. In ΔABC, AP : PB = 2 : 3. PO is parallel to BC and is extended to Q so that CQ is parallel to BA. The ratio area ΔAPO : area ΔABC is: [2008, 3 marks]
(a) 4 : 9
(b) 4 : 25
(c) 2 : 5
(d) 2 : 3
Answer: (b) 4 : 25

Q9. In the given figure, AB = 7 cm and BC = 9 cm. If ΔACD ~ ΔDCB, the length of CD is: [2009, 3 marks]
(a) 10 cm
(b) 11 cm
(c) 12 cm
(d) 13 cm
Answer: (c) 12 cm

Q10. In ΔABC, DE || BC and $PM/MQ = 2/3$. The ratio MN/QR is: [2018, 3 marks]
(a) 2 : 3

(b) 2 : 5
(c) 3 : 2
(d) 5 : 2
Answer: (b) 2 : 5

CHAPTER 14: CIRCLES

Q1. The circumcentre of a triangle is the point which is: [2024, 1 mark]
(a) at equal distance from the three sides of the triangle
(b) at equal distance from the three vertices of the triangle
(c) the point of intersection of the three medians
(d) the point of intersection of the three altitudes of the triangle
Answer: (b) at equal distance from the three vertices of the triangle

Q2. Statement 1: The point which is equidistant from three non-collinear points D, E and F is the circumcentre of ΔDEF.
Statement 2: The incentre of a triangle is the point where the bisector of the angles intersects. [2025 Paper, 1 mark]
(a) Both the statements are true
(b) Both the statements are false
(c) Statement 1 is true and Statement 2 is false
(d) Statement 1 is false and Statement 2 is true
Answer: (a) Both the statements are true

Q3. In the given diagram, chords AC and BC are equal. If ∠ACD = 120°, then ∠AEC is: [2025 Paper, 1 mark]
(a) 30°
(b) 60°
(c) 90°
(d) 120°
Answer: (d) 120°

Q4. ABCD is a cyclic quadrilateral. If ∠BAD = (2x + 5)° and ∠BCD = (x + 10)°, then x is equal to: [2022 Sem 2, 1 mark]
(a) 65°
(b) 45°
(c) 55°

(d) 5°
Answer: (c) 55°

Q5. In the given diagram RT is a tangent touching the circle at S. If ∠PST = 30° and ∠SPQ = 60°, then ∠PSQ is equal to: [2023, 1 mark]
(a) 40°
(b) 30°
(c) 60°
(d) 90°
Answer: (d) 90°

Q6. PQRS is a cyclic quadrilateral. Given ∠QPS = 73°, ∠PQS = 55° and ∠PSR = 82°, then ∠QRS is: [2018, 4 marks]
(a) 97°
(b) 107°
(c) 117°
(d) 127°
Answer: (b) 107°

Q7. In the given figure O is the centre of the circle, ∠BAD = 75° and chord BC = chord CD. Then ∠BOC is: [2009, 4 marks]
(a) 65°
(b) 70°
(c) 75°
(d) 80°
Answer: (c) 75°

Q8. AB and CD are two parallel chords of a circle such that AB = 24 cm and CD = 10 cm. If the radius of the circle is 13 cm, the distance between the two chords is: [2017, 4 marks]
(a) 15 cm
(b) 16 cm
(c) 17 cm
(d) 18 cm
Answer: (c) 17 cm

Q9. In the given figure, O is the centre of the circle. AB and CD are two chords of the circle. OM ⊥ AB, ON ⊥ CD, AB = 24 cm, OM = 5 cm, ON = 12 cm. The radius of the circle is: [2014, 4 marks]
(a) 10 cm

(b) 12 cm
(c) 13 cm
(d) 15 cm
Answer: (c) 13 cm

Q10. In the given figure, O is the centre of the circle and AB is a tangent at B. If AB = 15 cm and AC = 7.5 cm, the radius of the circle is: [2012, 3 marks]
(a) 7.5 cm
(b) 10 cm
(c) 11.25 cm
(d) 12.5 cm
Answer: (c) 11.25 cm

CHAPTER 15: TRIGONOMETRY

Q1. Statement 1: $\sin^2\theta + \cos^2\theta = 1$
Statement 2: $\text{cosec}^2\theta + \cot^2\theta = 1$
Which of the following is valid? [2024, 1 mark]
(a) only 1
(b) only 2
(c) both 1 and 2
(d) neither 1 nor 2
Answer: (a) only 1

Q2. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$
Reason (R): $1 - \sin^2 A = \cos^2 A$ [2025 Paper, 1 mark]
(a) (A) is true, (R) is false
(b) (A) is false, (R) is true
(c) Both (A) and (R) are true and (R) is the correct reason for (A)
(d) Both (A) and (R) are true and (R) is the incorrect reason for (A)
Answer: (c) Both (A) and (R) are true and (R) is the correct reason for (A)

Q3. $(1+\sin A)(1-\sin A)$ is equal to: [2023, 1 mark]
(a) $\text{cosec}^2 A$
(b) $\sin^2 A$
(c) $\sec^2 A$

(d) $\cos^2 A$
Answer: (d) $\cos^2 A$

Q4. Evaluate without using trigonometric tables: $(\sin 35^\circ \cos 55^\circ + \cos 35^\circ \sin 55^\circ) / (\operatorname{cosec}^2 10^\circ - \tan^2 80^\circ)$. The value is: [2010, 3 marks]
(a) 0
(b) 1
(c) 2
(d) 3
Answer: (b) 1

Q5. Prove that $\sin A/(1+\cos A) + (1+\cos A)/\sin A = 2\operatorname{cosec} A$. This is: [2009, 3 marks]
(a) True for all A
(b) True for A = 0°
(c) True for A = 90°
(d) True only for acute angles
Answer: (a) True for all A

Q6. Without using trigonometric tables evaluate: $\cos^2 26^\circ + \cos 64^\circ \sin 26^\circ + \tan 36^\circ / \cot 54^\circ$. The value is: [2012, 3 marks]
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (b) 2

Q7. Without using tables evaluate: $3\cos 80^\circ \operatorname{cosec} 10^\circ + 2\sin 59^\circ \sec 31^\circ$. The value is: [2013, 3 marks]
(a) 3
(b) 4
(c) 5
(d) 6
Answer: (c) 5

Q8. Without using trigonometric tables, evaluate: $\sin^2 28^\circ + \sin^2 62^\circ + \tan^2 38^\circ - \cot^2 52^\circ + \frac{1}{4} \sec^2 30^\circ$. The value is: [2017, 3 marks]
(a) 1
(b) 4/3
(c) 3/4

(d) 2
Answer: (b) 4/3

CHAPTER 16: HEIGHTS AND DISTANCES

Q1. A lighthouse is 80 m high. The angle of elevation of its top from a point 80 m away from its foot along the same horizontal line is: [2022 Sem 2, 1 mark]
(a) 60°
(b) 45°
(c) 30°
(d) 90°
Answer: (b) 45°

Q2. The angle of elevation of the top of a 100 m high tree from two points A and B on the opposite side of the tree are 52° and 45° respectively. The distance AB is: [2024, 5 marks]
(a) 178 m
(b) 180 m
(c) 182 m
(d) 185 m
Answer: (a) 178 m

Q3. From the top of a tower 100 m high, a man observes the angles of depression of two ships A and B on opposite sides of the tower as 45° and 38° respectively. The distance between the two ships is: [2019, 4 marks]
(a) 200 m
(b) 228 m
(c) 250 m
(d) 275 m
Answer: (b) 228 m

Q4. A man observes the angle of elevation of the top of a building to be 30°. He walks towards it in a horizontal line through its base. On covering 60 m, the angle of elevation changes to 60°. The height of the building is: [2011, 4 marks]
(a) 45 m
(b) 52 m
(c) 55 m

(d) 60 m
Answer: (b) 52 m

Q5. A vertical pole and a vertical tower are on the same level ground. From the top of the pole, the angle of elevation of the top of the tower is 60° and the angle of depression of the foot of the tower is 30°. If the height of the pole is 20 m, the height of the tower is: [2008, 4 marks]
(a) 60 m
(b) 70 m
(c) 80 m
(d) 90 m
Answer: (c) 80 m

Q6. From two points A and B on the same side of a building, the angles of elevation of the top of the building are 30° and 60° respectively. If the height of the building is 10 m, the distance between A and B is: [2009, 4 marks]
(a) 10.55 m
(b) 11.55 m
(c) 12.55 m
(d) 13.55 m
Answer: (b) 11.55 m

Q7. The angles of depression of two ships A and B as observed from the top of a lighthouse 60 m high are 60° and 45° respectively. If the two ships are on the opposite sides of the lighthouse, the distance between them is: [2017, 3 marks]
(a) 80 m
(b) 85 m
(c) 90 m
(d) 95 m
Answer: (d) 95 m

CHAPTER 17: MENSURATION (CYLINDER, CONE, SPHERE)

Q1. The circumference of the base of a cylindrical vessel is 132 cm and its height is 25 cm. The radius of the cylinder is: [2018, 2 marks]
(a) 21 cm
(b) 14 cm

(c) 7 cm
(d) 10.5 cm
Answer: (a) 21 cm

Q2. The volume of a conical tent is 462 m^3 and the area of the base is 154 m^2. The height of the cone is: [2022 Sem 2, 1 mark]
(a) 15 m
(b) 12 m
(c) 9 m
(d) 24 m
Answer: (c) 9 m

Q3. The radius of a roller 100 cm long is 14 cm. The curved surface area of the roller is (Take $\pi = 22/7$): [2022 Sem 2, 1 mark]
(a) 13200 cm^2
(b) 15400 cm^2
(c) 4400 cm^2
(d) 8800 cm^2
Answer: (d) 8800 cm^2

Q4. Volume of a cylinder of height 3 cm is 48π cm^3. Radius of the cylinder is: [2023, 1 mark]
(a) 48 cm
(b) 16 cm
(c) 4 cm
(d) 24 cm
Answer: (c) 4 cm

Q5. If the volume of two spheres is in the ratio 27 : 64, then the ratio of their radii is: [2025 Paper, 1 mark]
(a) 3 : 4
(b) 4 : 3
(c) 9 : 16
(d) 16 : 9
Answer: (a) 3 : 4

Q6. A solid metallic cylinder is cut into two identical halves along its height. The diameter of the cylinder is 7 cm and the height is 10 cm. The total surface area of both halves is: [2024, 4 marks]
(a) 437 cm^2

(b) 447 cm^2
(c) 457 cm^2
(d) 467 cm^2
Answer: (a) 437 cm^2

Q7. A solid metallic sphere of radius 6 cm is melted and made into a solid cylinder of height 32 cm. The radius of the cylinder is: [2019, 4 marks]
(a) 2 cm
(b) 3 cm
(c) 4 cm
(d) 5 cm
Answer: (b) 3 cm

Q8. A hollow sphere of external diameter 10 cm and internal diameter 6 cm is melted and made into a solid right circular cone of height 8 cm. The radius of the cone is: [2025 Paper, 3 marks]
(a) 5 cm
(b) 6 cm
(c) 7 cm
(d) 8 cm
Answer: (c) 7 cm

Q9. A solid cone of radius 5 cm and height 8 cm is melted and made into small spheres of radius 0.5 cm. The number of spheres formed is: [2011, 3 marks]
(a) 200
(b) 400
(c) 600
(d) 800
Answer: (b) 400

Q10. The volume of a conical tent is 1232 m^3 and the area of the base is 154 m^2. The height of the tent is: [2008, 4 marks]
(a) 20 m
(b) 22 m
(c) 24 m
(d) 26 m
Answer: (c) 24 m

CHAPTER 18: STATISTICS (MEAN, MEDIAN, MODE)

Q1. The median of the following observations arranged in ascending order is 64: 27, 31, 46, 52, x, x+4, 71, 79, 85, 90. The value of x is: [2024, 1 mark]
(a) 60
(b) 61
(c) 62
(d) 66
Answer: (c) 62

Q2. The median class for the given distribution is: [2022 Sem 2, 1 mark]

Class Interval	0-10	10-20	20-30	30-40
Frequency	2	4	3	5

(a) 0-10
(b) 10-20
(c) 20-30
(d) 30-40
Answer: (c) 20-30

Q3. The modal class of a given distribution always corresponds to the: [2022 Sem 2, 1 mark]
(a) interval with highest frequency
(b) interval with lowest frequency
(c) the first interval
(d) the last interval
Answer: (a) interval with highest frequency

Q4. Assertion (A): The mean of first 9 natural numbers is 4.5.
Reason (R): Mean = Sum of all the observations / Total number of observations [2025 Paper, 1 mark]
(a) (A) is true, (R) is false
(b) (A) is false, (R) is true
(c) Both (A) and (R) are true and (R) is the correct reason for (A)
(d) Both (A) and (R) are true and (R) is the incorrect reason for (A)
Answer: (b) (A) is false, (R) is true

Q5. Which of the following cannot be determined graphically for a grouped frequency distribution? [2023, 1 mark]
(a) Median
(b) Mode

(c) Quartiles
(d) Mean
Answer: (d) Mean

Q6. The mean of the following data is 16. The value of f is: [2020, 3 marks]

Marks	5	10	15	20	25
Students	3	7	f	9	6

(a) 4
(b) 5
(c) 6
(d) 7
Answer: (c) 6

Q7. The marks obtained by 30 students in a class assignment of 5 marks are given below:

Marks	0	1	2	3	4	5
No. of Students	3	6	10	5	5	1

The mean of the above distribution is: [2015, 3 marks]
(a) 2.5
(b) 2.6
(c) 2.7
(d) 2.8
Answer: (b) 2.6

Q8. The distribution given below shows the marks obtained by 25 students:

Marks	5	6	7	8	9	10
No. of Students	3	9	6	4	2	1

The median is: [2010, 4 marks]
(a) 6
(b) 6.5
(c) 7
(d) 7.5
Answer: (b) 6.5

Q9. The marks of 10 students in an examination arranged in ascending order are: 13, 35, 43, 46, x, x+4, 55, 61, 71, 80. If the median marks is 48, the value of x is: [2017, 4 marks]
(a) 44
(b) 45
(c) 46

(d) 47
Answer: (c) 46

Q10. The mean of the following distribution is 52 and the frequency of class interval 30-40 is 'f'. The value of f is: [2010, 4 marks]

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	5	3	f	7	2	6	13

(a) 4
(b) 5
(c) 6
(d) 7
Answer: (d) 7

CHAPTER 19: PROBABILITY

Q1. A die is thrown once. Assertion (A): The probability of getting an even number is 2/3. Reason (R): The sample space for even numbers on a die is {2, 4, 6}. [2024, 1 mark]
(a) A is true, R is false
(b) A is false, R is true
(c) Both A and R are true
(d) Both A and R are false
Answer: (b) A is false, R is true

Q2. Which of the following cannot be the probability of any event? [2025 Paper, 1 mark]
(a) 5/4
(b) 0.25
(c) 1/33
(d) 67%
Answer: (a) 5/4

Q3. The probability of getting a number divisible by 3 in throwing a dice is: [2022 Sem 2, 1 mark]
(a) 1/6
(b) 1/3
(c) 1/2
(d) 2/3
Answer: (b) 1/3

Q4. A letter is chosen at random from all the letters of the English alphabets. The probability that the letter chosen is a vowel is: [2023, 1 mark]
(a) 4/26
(b) 5/26
(c) 21/26
(d) 5/24
Answer: (b) 5/26

Q5. Cards bearing numbers 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 are kept in a bag. A card is drawn at random. The probability of getting a prime number is: [2018, 4 marks]
(a) 1/10
(b) 1/5
(c) 3/10
(d) 2/5
Answer: (a) 1/10

Q6. A game of numbers has cards marked with 11, 12, 13, ... 40. A card is drawn at random. The probability that the number on the card is a perfect square is: [2016, 3 marks]
(a) 1/10
(b) 2/15
(c) 3/20
(d) 4/25
Answer: (b) 2/15

Q7. Two coins are tossed once. The probability of getting at least 1 tail is: [2012, 3 marks]
(a) 1/4
(b) 1/2
(c) 3/4
(d) 1
Answer: (c) 3/4

Q8. A bag contains 5 white balls, 6 red balls and 9 green balls. A ball is drawn at random. The probability that the ball drawn is a white or a red ball is: [2015, 3 marks]
(a) 5/20
(b) 6/20
(c) 11/20
(d) 9/20
Answer: (c) 11/20

Q9. Sixteen cards are labeled as a, b, c, m, n, o, p. They are put in a box and shuffled. The probability that the card drawn is a vowel is: [2017, 4 marks]
(a) 1/4
(b) 3/4
(c) 1/2
(d) 5/8
Answer: (a) 1/4

Q10. From a pack of 52 playing cards, all cards whose numbers are multiples of 3 are removed. A card is now drawn at random. The probability of getting a face card is: [2011, 3 marks]
(a) 3/13
(b) 4/13
(c) 5/13
(d) 6/13
Answer: (a) 3/13

CHAPTER 20: DIRECT AND INVERSE VARIATION

Q1. The given table shows the distance covered and the time taken by a train moving at a uniform speed along a straight track:

Distance (in m)	60	90	y
Time (in s)	2	x	5

The values of x and y are: [2024, 1 mark]
(a) x = 4, y = 150
(b) x = 3, y = 100
(c) x = 4, y = 100
(d) x = 3, y = 150
Answer: (d) x = 3, y = 150